IN THE COURT OF APPEALS FOR THE STATE OF OHIO
TENTH APPELLATE DISTRICT

State of Ohio, ex. rel. Buckeye Environmental Network,
P.O. Box 82314
Columbus, Ohio 43202
Relator,

Case No. _____
Original Action in Mandamus
ORAL ARGUMENT REQUESTED

-vs-

Mary Mertz, Director, in her official capacity
Ohio Department of Natural Resources
2045 Morse Rd.
Columbus, OH 43229-6693
and
Eric Vendel, Chief, in his official capacity
Ohio Department of Natural Resources, Division of Oil and Gas Resource Management
2045 Morse Rd., Bldg. F2
Columbus, OH 43229-6693
Respondents.

VERIFIED PETITION FOR WRIT OF MANDAMUS

Relator Buckeye Environmental Network hereby submits, by and in the name of the State of Ohio, the following Petition for a Writ of Mandamus.

PRELIMINARY STATEMENT

1. Relator seeks a writ of mandamus to compel Respondents, officers within the Ohio Department of Natural Resources ("ODNR") and its Division of Oil and Gas Resources Management ("DOGRM"), to vacate two permits ("Permits") issued to DeepRock Disposal Solutions, LLC ("DeepRock") to drill two oil and gas waste disposal wells, named Stephan #1 and American Growers #4 (collectively "Proposed Wells") just outside of Marietta, Ohio. Relator seeks to compel ODNR and DOGRM to comply with their mandatory duty to apply

current Ohio laws and regulations in their review of the applications for the Proposed Wells. The Permits are attached hereto as Exhibits A and B.

2. Class II wells dispose of oil and gas waste by underground injection. The Federal Safe Drinking Water Act and Ohio's delegated laws and regulations at R.C. 1509.22 and Adm.Code 1501:9-3 govern permits for Class II wells.

3. Fracking and the production of oil and gas produce billions of barrels of oil and gas waste every year. In the Marcellus and Utica shale plays, most of the liquid oil and gas waste is disposed of in Ohio through Class II injection wells.

4. Ohio's regulations on Class II injection wells were not adequate to prevent serious environmental issues, including waste surfacing miles beyond injection sites and earthquakes linked to multiple injection wells throughout the state. ODNR has determined that some injection wells permitted under these regulations threaten water supplies, human health, and the environment. As a result, ODNR enacted more stringent regulations on January 13, 2022 that include numerous additional protections compared to the previous rules including: banning injection into some formations on a case-by-case basis, limiting the amount of injection under certain circumstances, stronger requirements for cementing wells, siting requirements from sensitive areas, stricter seismic monitoring, and more notice and public participation in the permitting process ("Current Rules" or "Current Regulations"). See Adm.Code 1501:9-3-05.

5. Under Ohio law, because a permit is a privilege and not a right, the filing of an application for a permit cannot create a vested right in obtaining that permit, and the laws and regulations in place at the time of granting the permit must govern. See *Scharff v. State*, 99 Ohio App. 139, 142 (1955).

6. Under Ohio's Class II regulations, DOGRM's review of the application for compliance with the law is not complete until after the application has been through the notice and comment period, including DOGRM's review and consideration of those comments and reaching a final permit decision. See Adm.Code 1501:9-3-05(E)(3)(c)(iii). Once the Current Regulations took effect January 13, 2022, DOGRM had a clear legal duty to apply them in considering applications.

7. In this case, DOGRM completed its technical review of the applications for the Proposed Wells in 2025—more than three years after the more protective regulations went into effect—giving both DOGRM and the applicant ample time to adjust to and apply the current regulations that took effect January 13, 2022. Yet, DOGRM abrogated its legal duty to apply the Current Regulations and instead applied a version of its rules, including its rules for permits to construct Class II wells, that were only in effect through January 12, 2022 (the "Old Rules" or "Old Regulations").

8. Accordingly, DOGRM issued final permits for the Proposed Wells using less protective areas of review, injection volumes, well construction requirements, and siting requirements, as well as more restrictive public participation procedures, than the Current Rules that have been in effect since January 13, 2022 require.

9. If constructed as permitted, DeepRock would develop the Proposed Wells in sensitive areas along the Ohio and Muskingum Rivers approximately two miles from two different Ohio municipal water systems—the City of Marietta and Warren Township—and less than two miles from Marietta's Source Water Protection Area.

10. As a result of DOGRM permitting the Proposed Wells without applying the Current Rules, people living, working, and recreating near the Proposed Wells will face increased risks to their health and safety.

11. These risks include exposure to toxic and harmful pollutants released into ground water, surface water, air, and soil from out of zone migration and accidents and spills on or near the injection sites. These risks also include seismic activity and potentially damaging earthquakes.

12. No other permitting process will consider these risks.

13. DOGRM acted unlawfully and abused its discretion when it issued the Permits for the Proposed Wells without applying the permitting rules in effect at the time of the application review, public notice, and permitting.

14. DOGRM's unlawful and unreasonable permitting of the Proposed Wells jeopardizes the health and safety of Relator's members; threatens their public water supply; threatens injuries to their property; harms their aesthetic, recreational, and environmental interests, and subjects them to unreasonable and unnecessary public health and safety risks.

15. Relator is a nonprofit organization with members and activities in Ohio that brings this action on behalf of its members, most of whom are Ohio residents and taxpayers, and who have a beneficial interest in ensuring DOGRM carries out its legal duties in permitting the Proposed Wells.

JURISDICTION AND VENUE

16. Jurisdiction lies with this court pursuant to Revised Code Chapter 2731, which governs mandamus proceedings and gives the court of appeals original jurisdiction over mandamus actions. This petition has been verified by affidavit in accordance with R.C. 2731.04.

17. The issuance of a permit to drill a new well "shall not be considered an order of the chief [of DOGRM]," and therefore cannot be appealed to the Oil and Gas Commission. See R.C. 1509.06(F); see also R.C. 1509.36 ("Any person adversely affected by an order by the chief of [DOGRM] may appeal to the oil and gas commission for an order vacating or modifying the order.").

18. Revised Code Chapter 119, which generally allows a party adversely affected by an order of an agency issued pursuant to an adjudication to seek judicial review of that order, does not apply to a permit to drill a new well. See R.C. 119.12(B), 1509.03(B)(1).

19. Therefore, Relator has no plain and adequate remedy at law to correct an abuse of discretion by DOGRM, overseen by ODNR, in the unlawful issuance of permits to drill the Proposed Wells, and their bringing of this petition for a writ of mandamus is appropriate.

20. The claims in this matter arise from Relator's clear legal rights and the clear legal duty of the Chief of DOGRM, overseen by the Director of ODNR, to uphold and carry out the statutory and regulatory requirements in place for the permitting of Class II wells.

21. Pursuant to Ohio Civ.R. 3(C), venue is proper because Respondents' principal offices are in Franklin County.

THE PARTIES

22. Buckeye Environmental Network ("BEN") is a non-profit organization headquartered in Columbus, Ohio with a mission of protecting communities and Ohio's environment from economic and environmental exploitation. BEN's mission includes addressing threats to human health and the environment from oil and gas extraction and waste disposal activities. BEN has members throughout the state of Ohio, including members who live, work, worship, and recreate

in Washington County, including in the City of Marietta and Warren Township, and near the location of the Proposed Wells.

23. BEN's members are Ohio taxpayers who have a beneficial interest in DOGRM following Ohio's laws regarding the permitting of Class II wells. Under R.C. 2731.02, BEN is a party beneficially interested in this matter, and has standing to commence this action.

24. Relator and/or its members filed timely, extensive written comments on the applications for the Proposed Wells, outlining numerous, substantial objections to the Wells.

25. Buckeye Environmental Network brings this petition on behalf of their members, including Dawn Hewitt, Dee Williams, and Betty Malcolm, who have completed sworn declarations at Exhibits C, D, and E of this Petition, respectively. The statements in the declarations are incorporated herein. The Proposed Wells threaten injuries to Relator's members' environment, property, water sources, aesthetics, and recreational activities.

26. Relator's members' threatened injuries are buttressed by the expert opinion and report of Dr. Catherine Helm-Clark, a professional geologist and geophysicist. Dr. Clark reviewed the application materials, the local geology, and oil and gas well and water well data, and concluded that DOGRM's permitting of the proposed wells threatens the environment and underground sources of drinking water in the areas where Relator's members live and recreate. The declaration of Dr. Clark and her report are attached hereto as Exhibit F. The statements and opinions in the declaration and attached report are incorporated herein.

27. Buckeye Environmental Network has members who reside in the City of Marietta and rely on the City's drinking water source as their primary water source. DOGRM's permitting of the Proposed Wells threatens these members' primary water source See Ex. C, ¶ 9 and Ex. D, ¶ ¶ 8–9.

28. Relator also has members that rely on Warren Township's drinking water source. DOGRM's permitting of the Proposed Wells threatens these members' primary water source. See Ex. E, ¶¶ 13, 15–16.

29. Buckeye Environmental Network has members who live and recreate in the area near the Proposed Wells. Relator's members are threatened with increased risks to the environment from the construction and operation of the proposed wells, including increased seismic activity, noise, increased truck traffic, the risk of accidents and spills, and the potential release of oil and gas waste into the environment, including the adjacent Muskingum and Ohio Rivers. See Ex. C, ¶ ¶ 10–18, Ex. D, ¶ ¶ 8, 10–17, and Ex. E, ¶ ¶ 12, 13–16.

30. Relator's members recreate in a park near the Proposed Wells, and boat on the Muskingum and Ohio Rivers in the area near the Proposed Wells, including the Ohio River Islands National Wildlife Refuge. See Ex. C, ¶ 11-12 and Ex. D, ¶ 11. The increased noise, seismic activity, increased truck traffic, and risk of accidents, releases, and spills, and the potential release of oil and gas waste into the environment from the permitting of the Proposed Wells will injure these members' aesthetic and recreational interests in the areas near the Proposed Wells. See Ex. C, ¶11–16, Ex. D, ¶ ¶ 11–16, and Ex. E, ¶ ¶ 12, 14. Relator's members would stop enjoying and recreating in these areas if the Proposed Wells are constructed and operated in accordance with the permits. See Ex. C, ¶ 11–18 and Ex. D, ¶ ¶ 11, 14.

31. DOGRM's failure to carry out the permitting requirements of the Federal Safe Drinking Water Act and Ohio Adm.Code Chapter 1501:9-3 has deprived Relator's members of the protections of the current law in Class II well permitting. The application of the current law is necessary to ensure the Proposed Wells do not result in out-of-zone migration; damaging

earthquakes; and contamination of underground sources of drinking water, the environment, and surface waters, including the Muskingum and Ohio Rivers. See Ex. F, ¶ ¶ 13–15.

32. A writ of mandamus revoking the Final Permits and mandating that DOGRM complete the legally required permitting process in the Federal Safe Drinking Water Act and Ohio Adm.Code Chapter 1501:9-3 would redress Relator's members' injuries. See Ex. C, ¶ ¶ 21–35, Ex. D, ¶ ¶ 20–26, 29–30, 34, and Ex. E, ¶ ¶ 19–33.

33. As Ohio citizens, Relator's members are interested in the execution of the laws in this state. DOGRM has created a rare and extraordinary case where an agency has refused to apply the clear requirements set forth in its own regulations when permitting activities that carry immense risk to public health, safety, and the environment.

34. Respondent Mary Mertz is sued in her official capacity as the Director of the Ohio Department of Natural Resources ("ODNR"). Respondent Mertz is required to ensure that all laws governing activities of the ODNR are faithfully executed.

35. Respondent Eric Vendel is sued in his official capacity as the Director of the Division of Oil and Gas Resource Management ("DOGRM") of ODNR. Chief Vendel oversees DOGRM staff involved in the permitting of Class II wells and is required to ensure that all laws governing DOGRM permitting of Class II wells are faithfully executed.

STATUTORY AND REGULATORY BACKGROUND

The Safe Drinking Water Act—Underground Injection Control Programs

36. Congress enacted the Safe Drinking Water Act ("SDWA"), 42 U.S.C. 300f et seq., to ensure the quality of the nation's drinking water and to protect it from contamination. As part of achieving these goals the SDWA includes, among other things, an underground injection control

("UIC") program that governs the permitting, operation, and closure of injection wells that place fluids underground for storage, disposal, or enhanced oil and gas recovery.

37. While the U.S. Environmental Protection Agency ("EPA") is ultimately responsible for administering the SDWA, Congress anticipated that the states could serve as the primary entities responsible for operating an underground injection control program. See 42 U.S.C. 300h–1(b)(2).

38. The SDWA establishes six classes of wells: Class I wells inject hazardous wastes, non-hazardous industrial wastes, or municipal wastewater; Class II wells inject brines and other fluids associated with oil and gas production and hydrocarbons for storage; Class III wells inject fluids associated with the solution mining of minerals; Class IV wells inject hazardous or radioactive wastes; Class V wells include all injection wells not covered in Classes I-IV; and Class VI wells inject carbon dioxide for long term storage. 40 C.F.R. 144.6.

39. Class II wells are used to inject fluids and wastes associated with oil and natural gas production. See *id.* at 144.6(b)(1).

40. Under the SDWA, a UIC well permit "does not convey any property rights of any sort, or any exclusive privilege." 40 C.F.R. 144.35(b). Moreover, a Class II well permit may be modified, reissued, or revoked when "standards or regulations on which the permit was based have been changed by promulgation of new or amended standards or regulations…" 40 C.F.R. 144.39(a)(3).

Ohio's Underground Injection Control Program for Class II Wells

41. Ohio has had primary enforcement authority over its underground injection control ("UIC") program for Class II wells under Section 1425 of the Safe Drinking Water Act since 1983.

42. R.C. 1509.22 requires DOGRM to adopt regulations "regarding the injection into wells of brine and other waste substances resulting from, obtained from, or produced in connection with oil or gas drilling, exploration, or production." R.C. 1509.22(D)(1).

43. The advent of hydraulic fracturing has caused an exponential increase in oil and gas waste over the last 15 years, with billions of tons of waste produced annually in the United States. This increase in waste production has caused a similar increase in the need for Class II disposal wells.

44. For the Marcellus and Utica shales, Ohio takes on the vast majority of the increase in oil and gas waste disposal through Class II wells, with much of the waste generated in Pennsylvania and West Virginia sent to Ohio wells for disposal.

45. Currently, Ohio has 232 active Class II injection wells, with approximately 19 more in the construction phase. Pennsylvania has approximately 16 Class II wells, and West Virginia has approximately 70 Class II wells.

Failures of Ohio's Class II Program Led to Updated Regulations

46. Since 2019, ODNR has found numerous incidents of oil gas waste migrating out of the injection zone from Class II injection wells. See Zuckerman, J., Ohio landowners say fracking wastewater is leaking underground, threatening their oil wells and drinking water, Signal (June 5, 2025), attached hereto as Exhibit G.

47. Oil and gas wastewater contains pollutants, toxic chemicals, and radioactive substances, including radium, strontium, selenium, thallium, lead, and other volatile organic compounds and heavy metals. See *id.*

48. Exposures to pollutants in oil and gas waste can be harmful to human health and safety and the environment.

49. In Washington County, ODNR determined that approximately 4.2 million gallons of Class II injected waste from the Redbird #4 injection well migrated to at least 28 production wells, with some wells being up to 5 miles away from the injection site. See City of Marietta Comments at Ex. 11 (Washington County Produced Water Investigation). A true and correct copy of the City's comments and exhibits is attached hereto as Exhibits H1, H2, H3 and H4.

50. DeepRock itself has caused out of zone migration issues with Class II wells permitted under the Old Regulations. In 2021, ODNR found that two Class II injection wells owned by DeepRock, the Warren Drilling Co. No. 1 Well ("Warren Well") and the Travis Unit No. 200405 Well ("Travis Well"), caused uncontrolled oil and gas waste to migrate to oil and gas production wells. The waste leaked for days onto the land and an adjacent stream, causing environmental contamination. See Ex. H2 at Ex. 6 (Chiefs Order 2023-02, ¶ 9).

51. The state incurred at least $1,279,608.03 in corrective action costs in responding to this migration incident. *Id.*

52. In 2023, these same DeepRock injection wells, the Warren Well and the Travis Well, again caused waste to migrate to production wells; this time the migration occurred more than 5 miles away from the injection sites. *Id.* at ¶ 10. The migration resulted in oil and gas waste spraying out of a production well, and ODNR determined this "caused imminent health, safety, and environmental risk." *Id.*

53. Owners of production wells in Washington County have sued injection well operators, including DeepRock, for causing oil and gas waste to migrate out of injection zones and into their production wells. See Exhibit G; see also *Bethel Oil & Gas, LLC v. Redbird Dev., LLC*, 2024-Ohio-5285, ¶ 2.

54. In 2023, ODNR determined that three injection wells in Athens County—permitted prior to the current regulations that took effect in 2022—caused waste to migrate out of the injection zone into production wells off site and impacted a production well that was 1.5 miles away from the injection wells. See Chiefs Order 2023-139. A true and correct copy of Chiefs Order 2023-139 is attached hereto as Exhibit I. ODNR determined that the out of zone migration from these wells "endanger and are likely to endanger public health, safety, or the environment" and "the continued operation of [the wells] presents an imminent danger to the health and safety of the public and is likely to result in immediate substantial damage to the natural resources of the state." *Id.* at ¶ 22.

55. Since the increase in Class II well activity, heightened seismic activity and earthquakes have been linked to Class II injection wells. See Buckeye Environmental Network Objection and Comment, at 9-1, a true and correct copy is attached hereto as Exhibit J.

Ohio's Current Permit to Construct Class II Well Laws

56. In response to waste migration events, seismic activity, and spills and accidents causing the release of waste into the environment, ODNR rewrote the regulations governing Class II wells.

57. On June 11, 2021, ODNR published a draft rewrite of the regulations for interested party review. A true and correct copy of the draft Class II disposal rules are attached hereto as Exhibit K.

58. The rewritten regulations went into effect on January 13, 2022, and remain in effect to the present day (hereinafter the "Current Regulations").

59. The Current Regulations, effective since January 13, 2022, include numerous provisions designed to be more protective of human health, the environment, and underground sources of drinking water, including:

a. A larger area of review to take corrective actions on nearby potential migration pathways;

b. Limitations or prohibitions on injection into certain formations;

c. Limitations on injection volumes where a well in the area of review penetrates the injection formation;

d. Stricter setbacks from sensitive areas, such as flood plains, public water sources, surface waters, and private dwellings;

e. Increased cement to be placed during the construction or conversion of a Class II well;

f. Enhanced testing of Class II wells during construction; and

g. Enhanced standards and requirements regarding seismic events near Class II wells.

60. Ohio Adm.Code 1501:9-3-03(A) states that "A person may only . . . dispose[] of brine or other waste substances at a Class II disposal well . . . in accordance with Chapter 1509. Of the Revised Code and rules adopted under it."

61. Further, "No person may inject brine and other waste substances or allow brine and other waste substances to migrate into an underground formation that is not approved for injection in the permit issued by the division." Adm.Code 1501:9-3-03(B).

62. R.C. 1509.05 states that "No person shall drill a new well . . . without having a permit to do so issued by the chief of the division of oil and gas resources management."

63. Current Ohio Adm.Code 1501:9-3-05 governs permits to construct Class II disposal wells.

64. Current Ohio Adm.Code 1501:9-3-05 states: "This rule applies to a new permit to construct a Class II disposal well and surface facility, a new permit to construct a Class II disposal well…"

65. DOGRM defines the "area of review" for Class II wells as "an area, established in rule 1501:9-3-05 of the Administrative Code, surrounding an existing or proposed Class II disposal well that is analyzed by the division in a process that consists of reviewing artificial penetrations of the injection zone, geological conditions, and potential migration pathways." Adm.Code 1501:9-3-01(E).

66. The current version of Ohio Adm.Code 1501:9-3-05 requires an area of review ("AOR") of 2 miles for wells injecting greater than an average volume of 1,000 barrels per day per year compared to only 0.5 miles for wells injecting more than 200 barrels per day required in the Old Regulations. See Adm.Code 1501:9-3-05(A)(1)(c).

67. Current Ohio Adm.Code 1501:9-3-05(B)(2) states:

> For a permit to drill a new Class II disposal well or to convert a well to a Class II disposal well, in which the injection formation will be located in a formation that has a producing well within the area of review either of the following apply:
>
> (a) The average disposal volume cannot exceed 200 barrels per day per year; or
>
> (b) The Class II disposal well owner must own each producing well in the area of review as long as the owner operates the Class II disposal well.

68. The Old Regulations did not restrict disposal volumes based on the location and producing formation for production wells within the area of review.

69. The Current Regulations include updated siting criteria, including that no portion of Class II well or surface facility may be located (1) within the boundary of a flood hazard area; (2) within five hundred feet from the boundary of the subject tract; (3) within one thousand feet of the five-year time of travel associated with a public drinking water supply or the emergency

management zone of a public water system intake; (4) within seven-hundred fifty feet of an occupied private dwelling or a public building; and (5) one hundred feet of a wetland or any surface water of the state. Adm.Code 1501:9-3-04(A)-(G).

70. The Current Regulations require enhanced public notice requirements, including direct notice of every application to well owners in the area of review, property owners within 1,500 feet of a proposed Class II well, and notice to the board of county commissioners and the board of township trustees where the Class II well is to be located. Adm.Code 1501:9-3-05(B)(3)(b)(i).

71. The Current Regulations require enhanced public participation, including a 30-day public comment period, and a public meeting for anyone receiving the enhanced notice described above. Adm.Code 1501:9-3-05(B)(3)(c).

DOGRM is Obligated to Apply the Current Law When Issuing a Permit

72. DOGRM's review of an application for a permit to drill is not complete until the applicant has publicly noticed the application for a permit to drill, the public comment period is fulfilled, and DOGRM has reviewed and considered the public comments and issued responses to comments. Prior to the completion of these legal obligations, DOGRM cannot issue a permit. See Adm.Code 1501:9-3-05(E).

73. DOGRM is required to consider whether an application meets the Class II rules through the public comment period and permit issuance. See *id.* at 1501:9-3-05(E)(3)(c)(iii).

74. Under Ohio law, there is no vested right in a permit application, and the law applicable at the time of making the final permit decision must govern. *Scharff*, 99 Ohio App. at 142.

75. Under the SDWA and Ohio regulations, a permit to drill a class II well does not convey a property interest or exclusive right. See 40 C.F.R. 144.35(b); Adm.Code 1501:9-3-09. Thus, an application for a permit cannot create a vested property interest or exclusive right.

76. ODNR is therefore required to apply the current Class II regulations to final decisions on permit applications occurring on or after January 13, 2022.

FACTUAL & PROCEDURAL BACKGROUND

Stephan #1

77. Stephan #1 is permitted to be located in Warren Township, just outside the city limits of Marietta, Ohio.

78. Stephan #1 is permitted to be located less than two miles from water wells that the City of Marietta relies on for its municipal water supply.

79. Stephan #1 is permitted to inject an average of 3,000 and a maximum of 5,000 barrels per day per year of oil and gas waste.

80. The Stephan #1 permit allows for multiple formations to be used as the injection zones, stating the injection zone is the "Clinton Sandstone through Medina Sandstone." Ex. A at 3.

81. DOGRM originally issued a permit to drill the Stephan #1 well on October 7, 2019. However, the permit expired because DeepRock did not drill the well within the legally mandated amount of time to commence drilling.

82. DOGRM received an application for a permit to "reissue" the Stephan #1 permit to drill from DeepRock on December 8, 2021.

83. DeepRock published a public notice of the Stephan #1 application on July 21, 2025, more than three years after the current rules were finalized.

84. A true and correct copy of the Stephan #1 application is attached and incorporated herein as Exhibit L.

85. The Stephan #1 application used a 0.5 mile-area of review.

86. Wilfong Unit #3 is a production well located approximately 1.22 miles away from Stephan #1.

87. DeepRock would have been required to identify Wilfong Unit #3 in its application for Stephan #1 if DeepRock had used a 2-mile area of review for its application for that well.

88. There are approximately 12 oil and gas wells located within a 0.5 mile radius of the proposed location of Stephan #1. Ex. F at Ex. 1, Table 5.

89. There are approximately 191 oil and gas wells located within a 2 mile radius of the proposed location of Stephan #1. Ex. F at Ex. 1, Table 5.

90. By applying the 0.5 mile area of review, DOGRM and DeepRock did not evaluate approximately 179 wells for corrective action that they would have evaluated for corrective action had they applied a 2 mile area of review for Stephan #1.

91. By applying the 0.5 mile area of review, DOGRM and DeepRock did not evaluate whether any of the approximately 179 wells triggered the need for a reduction or limitation in injection volumes for Stephan #1.

92. There are approximately 7 water wells located within a 0.5 mile radius of the proposed location of Stephan #1. Ex. F at Ex. 1, Table 1.

93. There are approximately 116 water wells located within a 2 mile radius of the proposed location of Stephan #1. Ex. F at Ex. 1, Table 1.

94. DOGRM held a 15-day comment period on the application for Stephan #1.

95. The current rules require a 30-day comment period. Adm.Code 1501:9-3-05(E)(3)(c)(i).

96. DeepRock did not provide notice of the Stephan #1 application to well owners within the area of review, landowners within 1,500 feet of the proposed well, or local township officials as required by Adm.Code 1501:9-3-05(E)(3)(b)(i).

97. Relator submitted comments objecting to Stephan #1 and requested a public hearing. In their comments, Relator raised numerous detailed objections related to technical deficiencies and public health and safety, including the unlawful application of the Old Rules.

98. The City of Marietta also submitted comments objecting to Stephan #1 and requested a public hearing. The City of Marietta's comments raised numerous detailed objections related to technical deficiencies and public health and safety, including the application of the Old Regulations, the location of the Proposed Wells in relation to public drinking water sources and other sensitive areas, injection migration and integrity issues with existing Class II wells in Washington County, and risks related to injecting in the local geology. See Ex. H1.

99. Despite having more than three years to consider the application under the current regulations, DOGRM did not apply the current regulations and reviewed the application for the Stephan #1 permit under the Old Regulations during its technical review and final permit issuance.

100. DOGRM approved the permit application for Stephan #1 and issued a final permit dated August 28, 2025 for that well. See Ex. A at 1.

101. The Permit for Stephan # 1 states "A PERMIT IS A PRIVILEDGE [sic] AND NOT A RIGHT, AND IS CONDITIONED ON COMPLIANCE WITH ALL APPLICABLE LAWS, THIS PERMIT, AND ORDERS ISSUED BY THE CHIEF." *Id.* at 6 (emphasis in original).

102. DOGRM published a Response to Comments on September 2, 2025. A true and correct copy of DOGRM's response to comments for Stephan #1 is attached and incorporated herein as Exhibit M.

103. In its Response to Comments on Stephan #1, DOGRM stated that it would not hold a public hearing. *Id* at 6.

104. In its Response to Comments on Stephan #1, DOGRM acknowledged that it had applied the Old Rules and had done so because it had received the Stephan #1 application prior to the effective date of the new rules. DOGRM did not provide any legal authority for this position, nor did it address the comments explaining that under Ohio law DORGM must apply the laws in place at the time the permit is issued. See *id.* at 1.

105. DOGRM subsequently published an undated document providing additional responses to concerns about both Proposed Wells. DOGRM's response document is attached and incorporated herein as Exhibit N.

106. In this subsequent response, DOGRM stated that it "does not have enough information to determine whether the Stephan #1 would be permitted under the new permitting rules." *Id.* at 2.

American Growers #4

107. American Growers #4 is permitted to be located in Warren Township, just outside the city limits of Marietta, Ohio.

108. American Growers #4 is permitted to inject an average of 3,000 and a maximum of 5,000 barrels per day per year of oil and gas waste.

109. The American Growers #4 permit allows multiple formations to be used as the injection zone, stating the injection zone is the "Oriskany Sandstone through Medina Sandstone." Ex. B at 3.

110. DOGRM received the application for a permit to drill American Growers #4 from DeepRock on or about December 7, 2021.

111. DeepRock published a public notice of the American Growers #4 application on January 30, 2025, more than three years after the effective date of the current rules.

112. A true and correct copy of the American Growers #4 application is attached and incorporated herein as Exhibit O.

113. The American Growers #4 application used a 0.5 mile area of review and identified one production well that is producing from the same formation, Wilfong Unit #3 (API #34167296620000).

114. Wilfong Unit #3 is owned and operated by Heinrich Enterprises.

115. Wilfong Unit #3 is located 0.56 miles away from American Growers #4.

116. Wilfong Unit #3 produces from the Medina formation.

117. Elkem-Price #3 (B-1), also named Virginia Trout Gustke, (API # 34167226510000) is a production well located approximately 1.87 miles away from American Growers #4.

118. Elkem-Price #3 (B-1) is owned and operated by Beardmore Oil & Gas Co.

119. Elkem-Price #3 (B-1) lists the Oriskany formation as a producing formation. A true and correct copy of ODNR's Well Summary Card for the Elkem Price #3 well is attached hereto as Exhibit P.

120. DeepRock would have been required to identify Elkem-Price #3 (B-1) in its application for American Growers #4 if DeepRock had used a 2-mile AOR for its application for that well.

121. DeepRock did not identify Elkem-Price #3 (B-1) in its application.

122. There are approximately 14 oil and gas wells located within a 0.5 mile radius of the proposed location of American Growers #4. Ex. F at Ex. 1, Table 7.

123. There are approximately 202 oil and gas wells located within a 2 mile radius of the proposed location of American Growers #4. Ex. F at Ex. 1, Table 7.

124. By applying the 0.5 mile area of review, DOGRM and DeepRock did not evaluate approximately 188 wells for corrective action that they would have evaluated for corrective action had they applied a 2 mile area of review.

125. By applying the 0.5 mile area of review, DOGRM and DeepRock did not evaluate whether any of approximately 188 wells triggered the need for a reduction or limitation in injection volumes for American Growers #4.

126. There are approximately 6 water wells located within a 0.5 mile radius of the proposed location of American Growers #4. Ex. F at Ex. 1, Table 1.

127. There are approximately 44 water wells located within a 2 mile radius of the proposed location of American Growers #4. Ex. F at Ex. 1, Table 1.

128. DOGRM held a 15-day comment period on the application for American Growers #4.

129. The current rules require a 30-day comment period. Adm.Code 1501:9-3-05(E)(3)(c)(i).

130. DeepRock did not provide notice of the American Growers #4 application to well owners within the area of review, landowners within 1,500 feet of the proposed well, or local township officials as required by Adm.Code 1501:9-3-05(E)(3)(b)(i).

131. DOGRM did not apply the current regulations and reviewed the application for the American Growers #4 permit under the Old Regulations during its technical review.

132. DOGRM approved the permit application for American Growers #4 and issued a final permit for the well on March 7, 2025. See Ex. B at 1.

133. The Permit for American Growers # 4 states "A PERMIT IS A PRIVILEDGE [sic] AND NOT A RIGHT, AND IS CONDITIONED ON COMPLIANCE WITH ALL APPLICABLE LAWS, THIS PERMIT, AND ORDERS ISSUED BY THE CHIEF." *Id.* at 6 (emphasis in original).

134. DOGRM published a Response to Comments on March 11, 2025. A true and correct copy of DOGRM's response to comments for American Growers #4 is attached as Exhibit Q.

135. Relator asked DOGRM multiple times about why DOGRM was not applying its rules, which became effective in 2022, to the permits for the Proposed Wells.

136. In its Response to Comments on American Growers #4, DOGRM did not address why it was applying the outdated regulations to the permit application. See Ex. Q.

137. DeepRock has indicated to DOGRM its intent to pull its application for American Growers #4 and relocate the well because of spacing issues with other wells. See June 30, 2025 email from DeepRock consultant to Ohio EPA, attached hereto as Exhibit R.

138. As of the date of this filing, Relator has not received notice that DOGRM has vacated the permit for American Growers #4.

Risks to Public Health and the Environment

139. The Proposed Wells would be located in Warren Township, just outside the boundaries of Marietta, Ohio, and near many sensitive areas and potential risks including the Muskingum and Ohio Rivers, well fields for public water sources, private water wells, and abandoned and producing oil and gas wells. See Ex. H2 at Ex. 3.

140. Marietta has a population of approximately 13,000 people.

141. The Proposed Wells are located within two miles of Marietta's Source Water Protection Area. Ex. H2 at Ex. 4.

142. America Grower's #4 is proposed to be located approximately 10,595 feet from Marietta's Source Water Protection Area. Ex. H2 at Ex. 4.

143. Stephan #1 is proposed to be located approximately 9,050 feet from Marietta's Source Water Protection Area. Ex. F at Ex. 1, Table 1.

144. Marietta sources its drinking water from seven groundwater wells located in a sand and gravel aquifer.

145. Stephan #1 is proposed to be located less than two miles from wells that the City of Marietta relies on for its municipal water supply. Ex. F at Ex. 1, Table 1.

146. Marietta's public drinking water system is a source of drinking water for approximately 18,961 people.

147. Warren Township in Washington County, Ohio has a population of approximately 4,000 people.

148. The Warren Community Water and Sewer Association provides Warren Township residents with drinking water from a well field on the west side of the Muskingum River, approximately two miles away from the Proposed Wells. See Ex. H2 at Ex. 3.

149. Excluding the Proposed Wells, four other Class II injection wells are located within two miles of the City of Marietta's Source Water Protection Area. Ex. H2 at 3.

150. The four Class II injection wells currently operating within two miles of Marietta's Source Water Protection Area are collectively permitted to inject up to 20,000 barrels per day into the Medina Sandstone. Ex. H2 at 3.

151. When several injection wells are near one another, the cumulative pressure in underground formations can be substantially higher than one individual well pressure. Ex. H4 at Ex. 23.

152. Volume, pressure, and rate of injection can impact the ability of rock to resist stress and cause breaks in the confining layer. Ex. H4 at Ex. 23.

153. All oil and gas wells in the 2-mile radii around the Proposed Wells are potential conduits for Class II waste to reach the surface or underground sources of drinking water ("USDWs") if out of zone migration of waste occurs. Ex. F at Ex. 1, 3-5, 34, 36.

154. Abandoned and/or improperly plugged wells present an even greater risk of contamination of USDWs and the environment than producing oil and gas wells. Ex. F at Ex. 1, p. 3, 17–18, 34; Ex. H4 at Ex. 23.

155. If operated as authorized by the Permits, the Proposed Wells will cause oil and gas waste to migrate outside of the intended injection formations and confining layers. Ex. F. at Ex. 1, p. 27-35; 40-41.

156. Once oil and gas waste has migrated outside of the intended injection formations and confining layers it can migrate to the surface and underground sources of drinking water. Ex. F. at Ex. 1, p. 37; 40-41.

157. The Proposed Wells would be located less than two miles from the Ohio River Islands National Wildlife Refuge.

158. The lateral continuity of geologic layers at depth in the region of the Proposed Wells is challenging to predict. Ex. H4 at Ex. 23.

159. Recently the region of the Proposed Wells has experienced an increase in earthquakes. Ex. H4 at Ex. 23.

160. The permit applications for the Proposed Wells fail to include any seismic or structural analysis to investigate the possibility of undiscovered buried faults.

161. The permit applications for the Proposed Wells fail to include any information addressing the integrity of the confining units over the injection intervals.

162. The target injection formations of the Proposed Wells include formations that are too tight for practical brine disposal, increasing the risk that pressures used in injection will create fractures that lead to oil and gas waste migrating outside of the target injection formations and into the environment, including into underground sources of drinking water and surface waters.

163. The volume of waste permitted to be injected by the Proposed Wells will cause fractures and migration pathways in the target formations and confining formations, and allow waste to leave the injection zone and threaten underground sources of drinking water and the environment. Ex. F at 35, 40-41.

164. The Proposed Wells, as permitted, increase the risk of contaminants and pollutants migrating outside of the targeted injection zone, to the surface and into the environment.

165. If waste injected into the Proposed Wells moves into upper geologic layers or underground sources of drinking water, the damage cannot be undone. Ex. H4 at Ex. 23.

166. As permitted, the Proposed Wells threaten underground sources of drinking water, including the aquifer that the City of Marietta relies on to supply its drinking water.

167. As permitted, the Proposed Wells threaten public health and safety and the environment.

COUNT 1

The Court should issue a Writ of Mandamus directing Respondents to vacate the permit to drill the Stephan #1 well because Respondents had a mandatory duty to apply current regulations for Class II injection wells to the application review and final permit issuance.

168. Relator restates and reiterates all preceding paragraphs of this petition as if fully rewritten herein and additionally alleges the following:

169. DOGRM's Current Regulations for permitting Class II wells came into effect on January 13, 2022.

170. DOGRM applied the Old Regulations to the application review and permit issuance, and issued DeepRock the permit for Stephan #1 on August 28, 2025.

171. DOGRM failed in its mandatory duty to apply current Ohio regulations to the Stephan 1 Class II well permit review and issuance, including, but not limited to:

a. Failing to require a 2-mile area of review for the application pursuant to Ohio Adm.Code 1501:9-3-05(A)(1)(c);

b. Failing to limit the average disposal volume to 200 barrels per day pursuant to Ohio Adm.Code 1501:9-3-05(B)(2)(a);

c. Failing to evaluate the application to ensure compliance with the siting criteria pursuant to Ohio Adm.Code 1501:9-3-04(A)-(G); and

d. Failing to follow the notice and public participation procedures pursuant to Ohio Adm.Code 1501:9-3-05(E)(3)(c), including failing to allow a 30-day public comment period.

172. Relator and its members would reasonably and foreseeably benefit from DOGRM following its own regulations and applying current protections in reviewing DeepRock's application for Stephan #1.

173. DOGRM's decision not to apply its own regulations deprived Relator and its members of protections and public participation opportunities to which they are entitled under the Safe Drinking Water Act and Ohio's implementing statutes and regulations.

174. DOGRM abused its discretion and acted unreasonably and arbitrarily when it did not apply the Current Regulations in reviewing the application and issuing the permit for the Stephan #1 well.

COUNT 2

The Court should issue a Writ of Mandamus directing Respondents to vacate the permit to drill the American Growers #4 well because Respondents had a mandatory duty to apply current regulations for Class II injection wells to the application review and final permit issuance.

175. Relator restates and reiterates all preceding paragraphs of this petition as if fully rewritten herein and additionally alleges the following:

176. DOGRM's Current Regulations for permitting Class II wells came into effect on January 13, 2022.

177. DOGRM applied the Old Regulations to the application review and permit issuance and issued DeepRock the permit for American Growers #4 on March 7, 2025.

178. DOGRM failed in its mandatory duty to apply current Ohio regulations to the American Growers #4 Class II well permit review and issuance, including, but not limited to:

a. Failing to require a 2-mile area of review for the application pursuant to Ohio Adm.Code 1501:9-3-05(A)(1)(c);

b. Failing to limit the average disposal volume to 200 barrels per day pursuant to Ohio Adm.Code 1501:9-3-05(B)(2)(a);

c. Failing to evaluate the application to ensure compliance with the siting criteria pursuant to Ohio Adm.Code 1501:9-3-04(A)-(G); and

d. Failing to follow the notice and public participation procedures pursuant to Ohio Adm.Code 1501:9-3-05(E)(3)(c), including failing to allow a 30-day public comment period.

179. Relator and its members would reasonably and foreseeably benefit from DOGRM following its own regulations and applying the current protections in reviewing the application.

180. DOGRM's decision not to apply its own regulations has deprived Relator and its members of protections and public participation opportunities to which they are entitled under the SDWA and Ohio's implementing statutes and regulations.

181. DOGRM abused its discretion and acted unreasonably and arbitrarily when it did not apply the Current Regulations in reviewing the application and approving the permit for the American Growers #4 well.

COUNT 3

The Court should issue a Writ of Mandamus directing Respondents to vacate the permits to drill for the Proposed Wells because Respondents had a mandatory duty to deny an application that threatens USDWs, human health, safety, and the environment.

182. Relator restates and reiterates all preceding paragraphs of this petition as if fully rewritten herein and additionally alleges the following:

183. The Proposed Wells "could reasonably be anticipated to cause damage or injury to public health or safety or the environment" in violation of R.C. 1509.22(A).

184. The permitting of the Proposed Wells violates the statutory mandate that "the chief shall not issue a permit for the injection of brine or other waste substances resulting from, obtained from, or produced in connection with oil or gas drilling, exploration, or production unless the chief concludes that the applicant has demonstrated that the injection will not result in the presence of any contaminant in ground water that supplies or can reasonably be expected to supply any public water system, such that the presence of the contaminant may result in the system's not complying with any national primary drinking water regulation or may otherwise adversely affect the health of persons." R.C. 1509.22(D)(3).

185. By failing to apply the Current Regulations, DOGRM abdicated its clear legal duty to ensure Class II wells are "designed, constructed, and operated in a manner that protects public health and safety and the environment." See Adm.Code 1501:9-3-05.

186. DOGRM violated Ohio Adm.Code 1501:9-3-05(A)(3) by approving the application without requiring corrective action on wells in the area of review that have limited or no records. This violation threatens USDWs and public health.

187. Relator and its members would reasonably and foreseeably benefit from DOGRM following its own regulations and ensuring the wells are permitted using the Current Regulations that are more protective of human health and safety, underground sources of drinking water, and the environment.

188. DOGRM's decision not to apply its own regulations has deprived Relator and its members of protections to which they are entitled under the SDWA and Ohio's implementing statutes and regulations.

189. Thus, DOGRM abused its discretion and acted unreasonably and arbitrarily when it issued the permits for the Proposed Wells.

WHEREFORE, Relator prays the Court issue a writ of mandamus under R.C. Ch. 2731 ordering DOGRM to:

(i) Vacate the final permit to drill American Growers #4;

(ii) Vacate the final permit to drill Stephan #1;

(iii) Apply the current Ohio Class II well regulations to the applications and permits for American Growers #4 and Stephan #1 wells; and

(iv) Reject the applications for permits to drill American Growers #4 and Stephan #1 as incomplete under Ohio's Class II well regulations.

Relator requests to be rewarded its costs and such other relief at law or equity as the Court may deem necessary and proper.

Respectfully submitted,

EARTHJUSTICE

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Counsel for Relator, Buckeye Environmental Network

VERIFICATION

I, Becca Pollard, am Executive Director of Buckeye Environmental Network ("BEN") and am empowered to bring the foregoing Verified Petition for a Writ of Mandamus on behalf of BEN and BEN's members. I have reviewed the allegations in the Petition and believe them to be true to the best of my knowledge, information, and belief.

[handwritten signature] Becca Pollard

State of Ohio) County of Franklin) ss:

Sworn to before me and subscribed in my presence this 5th day of November, 2025.

[handwritten signature] Notary Public

Matthew Allen Puterbaugh, Notary Public, State of Ohio, My Comm. Expires August 7, 2029. [notary seal]

INDEX OF EXHIBITS

Exhibit A: Stephan No. 1 Permit

Exhibit B: American Growers No. 4 Permit

Exhibit C: Declaration of Dawn Hewitt

Exhibit D: Declaration of Dee Arnold

Exhibit E: Declaration of Betty Malcolm

Exhibit F: Declaration of Catherine M. Helm-Clark w/ Exhibits 1 & 2

Exhibit G: Zuckerman, J., Ohio landowners say fracking wastewater is leaking underground, threatening their oil wells and drinking water, Signal (June 5, 2025)

Exhibit H1: City of Marietta Comments and Exhibits 1-1A

Exhibit H2: City of Marietta Exhibits 2-10A

Exhibit H3: City of Marietta Exhibits 11-18

Exhibit H4: City of Marietta Exhibits 19-23

Exhibit I: ODNR Chief Order No. 2023-139 (June 26, 2023)

Exhibit J: Buckeye Environmental Network Objection & Comment of Stephan #1 Well

Exhibit K: ODNR Draft Rewrite of Regulations for Interested Party Review

Exhibit L: DeepRock Disposal Solutions, LLC Permit Application for Stephan #1 Well (Dec. 8, 2021)

Exhibit M: DOGRM Response to Comments of Stephan #1 Well

Exhibit N: DOGRM Response to Washington City Officials

Exhibit O: DeepRock Disposal Solutions, LLC Permit Application for American Growers #4 Well

Exhibit P: ODNR's Well Summary Card for the Elkem Price #3

Exhibit Q: DOGRM Response to Comments for American Growers #4 (Mar. 11, 2025)

Exhibit R: Email from DeepRock Consultant to OH EPA (June 30, 2025)